Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	*(a)*	(i)	Increase pd until microammeter reads zero. **Or** decrease pd until microammeter just starts to read a current (1) Reading of voltmeter gives the energy value needed or equivalent statement (1) Accept $E_k = eV_{stop}$	2			2		2
		(ii)	$\phi = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^{8}}{405 \times 10^{-9}} [\mathrm{J}](1) - 0.480 \times 1.60 \times 10^{-19} [\mathrm{J}](1)$ [4.91×10^{-19}] [-7.68×10^{-20}] **or** $\phi = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^{8}}{405 \times 10^{-9} \times 1.60 \times 10^{-19}} [\mathrm{eV}](1) - 0.480 [\mathrm{eV}](1)$ [3.07] even if not transposed 4.1×10^{-19}[J] **or** 2.6 [eV] (1)		3		3	3	3
		(iii)	For 650 nm, photon energy = 3.1×10^{-19} J **or** 1.9 eV (1) Photon energy not enough to knock out electron **or** photon energy < ϕ (1) More photons [per second] won't help or increasing intensity doesn't increase photon energy (1) Accept only gives more photons at the same energy or doesn't change frequency **Alternative** Calculation of $f_0 = 6.2 \times 10^{14}$ [Hz] **ecf** or calculation of $\lambda_0 = 4.8 \times 10^{-7}$ [m] **ecf** (1) 4.6×10^{14} [Hz] < 6.2×10^{14} [Hz] **or** 650 nm is > 480 nm so no emission occurs (1) More photons [per second] won't help or increasing intensity doesn't increase photon energy (1) Accept only gives more photons at the same energy or doesn't change frequency - Subtract 1 mark if no useful reference made to photons			3	3		3

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Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)	$v = \frac{h}{m\lambda}$ transposed and used (1) $v = 1.5 \times 10^7$ [m s^{-1}] (1)		2		2	2	
		(ii)	Electrons are wavelike **or** diffract **or** interfere (1) Accept reference to wave-particle duality Graphite crystals act as diffraction gratings **or** electrons diffract <u>between (around) atom</u>[s] (through lattice) **or** λ comparable with gaps between atoms (1)	2			2		
			Question 7 total	**4**	**5**	**3**	**12**	**5**	**8**

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